$$\int Pdx = \int -3\cot x dx = -3\log \sin x$$

$$\therefore = -\log \sin^3 x = \log \frac{1}{\sin^3 x}$$

$$\text{I.F.} = e^{\log \frac{1}{\sin^3 x}} = \frac{1}{\sin^3 x}$$

The required solution is

$$y\,(I.F) = \int Q(I.F)dx + c$$

$$y\frac{1}{\sin^3 x} = \int \sin 2x \frac{1}{\sin^3 x}\, dx + c$$

$$y\frac{1}{\sin^3 x} = \int 2\sin x \cos x \times \frac{1}{\sin^3 x} dx + c$$

$$= 2\int \frac{1}{\sin x} \times \frac{\cos x}{\sin x}\, dx + c$$

$$= 2\int \operatorname{cosec} x \cot x dx + c$$

$$y\frac{1}{\sin^3 x} = -2\operatorname{cosec} x + c \qquad (1)$$

Now y = 2 when $x = \frac{\pi}{2}$

$$(1) \Rightarrow 2\left(\frac{1}{1}\right) = -2\times 1 + c \Rightarrow c = 4$$

$$\therefore (1) \Rightarrow y\frac{1}{\sin^3 x} = -2\operatorname{cosec} x + 4$$

**Example 4.24**

A firm has found that the cost C of producing $x$ tons of certain product by the equation $x\frac{dC}{dx} = \frac{3}{x} - C$ and C = 2 when x = 1.

Find the relationship between C and x.

***Solution:***

$$x\frac{dC}{dx} = \frac{3}{x} - C$$

$$\frac{dC}{dx} = \frac{3}{x^2} - \frac{C}{x}$$

$$\frac{dC}{dx} + \frac{C}{x} = \frac{3}{x^2}$$

i.e., $$\frac{dC}{dx} + \frac{1}{x}C = \frac{3}{x^2}$$

It is of the form $\frac{dC}{dx} + PC = Q$

Here, $P = \frac{1}{x}, Q = \frac{3}{x^2}$

$$\int Pdx = \int \frac{1}{x} dx = \log x$$

$$\text{I.F} = e^{\int pdx} = e^{\log x} = x$$

The Solution is

$$C(\text{I.F}) = \int Q(I.F)dx + k \quad \text{where k is constant}$$

$$Cx = \int \frac{3}{x^2} x dx + k$$

$$= 3\int \frac{1}{x} dx + k$$

$$Cx = 3\log x + k \qquad (1)$$

Given C = 2 When $x = 1$

$$(1) \Rightarrow 2\times 1 = k \Rightarrow k = 2$$

$\therefore$ The relationship between C and x is

$$Cx = 3\log x + 2$$

## Exercise 4.4

Solve the following:

1. $\frac{dy}{dx} - \frac{y}{x} = x$
2. $\frac{dy}{dx} + y\cos x = \sin x \cos x$
3. $x\frac{dy}{dx} + 2y = x^4$
4. $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{1+x^2}{1+x^3}$
5. $\frac{dy}{dx} + \frac{y}{x} = xe^x$
6. $\frac{dy}{dx} + y\tan x = \cos^3 x$
7. If $\frac{dy}{dx} + 2y\tan x = \sin x$ and if $y = 0$ when $x = \frac{\pi}{3}$ express $y$ in terms of $x$



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